

Name: _____

Date: _____

DIRECTIONS

This assessment covers lessons 4.04.01–4.04.06.

Show your work in the box. Write your final answer on the line.

Read every problem carefully. Some problems have more than one part — answer ALL parts.

Take your time. Check your work before you turn in.

1 Mental math fluency

Find each in your head.

(a) 25×4 (pattern) (b) 24×5 (decompose) (c) 99×7 (compensate)

SHOW YOUR MENTAL STEPS

(a) ____ (b) ____ (c) ____

2 Multiply by a multiple of 10

Find each product.

(a) 87×50 (b) 145×40 (c) 60×90 (d) 230×50

BASIC FACT $\times$ ZEROS

(a) ____ (b) ____ (c) ____ (d) ____

3 Estimate two ways

For $52,891 \times 4$:

- (a) Round to the nearest TEN-THOUSAND, then multiply.
(b) Round to the nearest THOUSAND, then multiply.
(c) Find the exact. (d) Which estimate was closer?

SHOW BOTH ESTIMATES AND THE EXACT

(a) ____ (b) ____ (c) ____ (d) ____

4 5-digit $\times$ 1-digit fluency

Find each product.

- (a) $23,475 \times 3$ (b) $41,629 \times 5$ (c) $19,256 \times 8$

STACK AND MULTIPLY

(a) ____ (b) ____ (c) ____

5 Partial products — 2-digit

Split the second factor into tens and ones. Show both partial products.

(a) 34×28 (b) 245×32 (c) 638×48

SHOW BOTH PARTIAL PRODUCTS

(a) ____ (b) ____ (c) ____

6 Standard algorithm — 2-digit

Find each using the standard algorithm. Stack the two partial rows.

(a) 47×35 (b) 356×47 (c) 824×63

STACK, SHIFT, AND ADD

(a) ____ (b) ____ (c) ____

7 Word problem — Atlas's library

Atlas counts 1,348 books on each of the library's 6 floors.

(a) Estimate by rounding to the nearest thousand. (b) Find the exact total.

ESTIMATE FIRST, THEN EXACT

(a) $\approx$ ____ (b) Exact: ____

8 Word problem — Hazel's seating

The auditorium has 28 rows of 36 seats each.

(a) Estimate. (b) Find the exact total.

SPLIT 36 INTO $30 + 6$

(a) $\approx$ ____ (b) Exact: ____

9 MODEL — area model with shift

Draw the partial-product model for 84×36 .

Show: $84 \times 6 = \underline{\quad}$, $84 \times 30 = \underline{\quad}$, Sum = $\underline{\quad}$.

Explain why 84×30 STARTS in the tens place, not the ones place.

DRAW THE TWO ROWS WITH THE SHIFT

Total: $\underline{\quad}$

10 EXPLAIN — Unit 4 reflection

Across Unit 4 you learned mental math (patterns, decompose, compensate),

the standard algorithm for 1-digit and 2-digit factors, multiplying by multiples of 10, and partial products.

In one paragraph, explain how knowing the partial-product / area model helps you understand WHY the standard algorithm works.

WRITE YOUR REFLECTION

My explanation: